Do Memory Tools Help Agents? The Surprising Failure of Dense Retrieval

Anonymous Authors
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Abstract

Memory systems for AI agents are typically evaluated on static retrieval benchmarks, where dense embeddings consistently outperform keyword search. We introduce **LME+**, an agentic adaptation of the LongMemEval benchmark, and evaluate five memory approaches: Oracle (perfect retrieval), frequency-based keyword search, dense embeddings (Stella V5), filesystem access, and a hybrid combining keyword + embeddings. Across 50 questions with a ReAct agent, **keyword search (62% [47, 75]) vastly outperformed dense embeddings (26% [15, 40])** and even their hybrid combination (42% [28, 57]), despite dense embeddings excelling on the static LME benchmark (71%). We identify a 28-point gap between Oracle (90%) and the best practical method, attributing this to retrieval quality rather than agent architecture. Our findings challenge the assumption that static retrieval performance translates to agentic settings and suggest frequency-based keyword search outperforms sophisticated dense retrieval for conversational memory in this setting. We discuss limitations including sample size, implementation choices, and generalizability.

1 Introduction

AI agents increasingly require long-term memory to maintain context across extended interactions. Production systems like ChatGPT [10] and Claude now support multi-session memory, while frameworks like LangChain [5] and Letta [7] provide memory primitives for agent developers. A critical question emerges: **how should we evaluate memory systems for agents?**

Current benchmarks evaluate memory systems on *static retrieval tasks*: given a query, retrieve the top-k most relevant documents [15, 13]. Dense embedding models like Stella V5 [18] consistently outperform sparse keyword search (BM25) on these benchmarks. However, agents use memory *dynamically*—issuing multiple queries, reformulating based on results, and iterating until success or timeout. **Do static retrieval gains translate to agentic gains?**

We introduce **LME+**, where a ReAct agent [17] actively queries memory tools to answer questions from LongMemEval [15]. We evaluate five approaches across 50 questions, measuring end-to-end QA accuracy, cost, and latency.

Our contributions are:

- **LME+ benchmark**: An agentic adaptation of LongMemEval for evaluating memory systems in realistic agent settings
- **Surprising negative results**: Dense embeddings (Stella V5) achieve only 26% accuracy, dramatically underperforming keyword search's 62%
- **Hybrid failure**: Combining keyword search + embedding reranking (42%) performs worse than keyword alone, suggesting embeddings actively demote correct results
- **Retrieval bottleneck**: A 28-point gap between Oracle (90%) and keyword search (62%) identifies retrieval quality, not agent architecture, as the primary limitation

2 Related Work

Memory Benchmarks for Agents. LongMemEval [15] evaluates retrieval systems on conversational QA, finding Stella V5 dense embeddings achieve 71% accuracy with top-5 retrieval in static settings. LoCoMo [9] presents very long-term conversations (300 turns, 9K tokens) testing information extraction, temporal reasoning, and multi-session reasoning. MemoryAgentBench [2] provides a unified benchmark evaluating four core memory competencies: accurate retrieval, test-time learning, long-range understanding, and conflict resolution. Letta's evaluation [7] found that gpt-4o-mini with file-based context management achieves 74% on LoCoMo's structured extraction tasks, suggesting memory management strategies matter beyond retrieval mechanisms alone. Our work tests whether static retrieval performance translates to agentic settings where agents iteratively query memory tools.

Dense vs Sparse Retrieval. Dense retrievers [3, 4] typically outperform BM25 [12] on static benchmarks [13]. Stella V5 [18] achieves state-of-the-art performance on MTEB. Recent work on hybrid retrieval [1] combines dense and sparse methods through fusion techniques like Reciprocal Rank Fusion (RRF) and convex combination, with convex fusion often outperforming cascade approaches. Our hybrid implementation uses cascade reranking; alternative fusion strategies may yield different results. Our work shows the dense-vs-sparse ranking reverses when retrieval is agentic.

Agent Memory Systems. MemGPT [11] uses hierarchical memory with embedding-based retrieval. Liu et al. [8] provide a comprehensive survey organizing memory systems across forms (token-level, parametric, latent), functions (factual, experiential, working), and dynamics (formation, evolution, retrieval). A-MEM [16] introduces dynamic memory organization using Zettelkasten principles, automatically linking related memories with structured attributes. Recent systems like MemR3 [14] use reflective reasoning to decide when to retrieve from long-term memory, while Hindsight [6] proposes retain/recall/reflect mechanisms for very long conversations. Our work empirically evaluates whether sophisticated retrieval mechanisms (dense embeddings, hybrid approaches) outperform simple keyword search when used by agentic systems.

3 LME+: Agentic Memory Benchmark

3.1 Task Formulation

Static LME [15]: Given conversation history $\mathcal{H} = \{s_1, \ldots, s_N\}$ and question q, retrieve top-k sessions $\{s_{i_1}, \ldots, s_{i_k}\}$ and answer q.

Agentic LME+: A ReAct agent [17] with memory tool T iteratively queries to answer q. The agent may:

- Issue query q' to tool: $T(q') \rightarrow$ sessions
- Reformulate based on results
- Iterate up to $I_{\max} = 5$ turns

3.2 Memory Adapters

All adapters expose a single `search_memory(query)` tool:

Oracle. Returns gold answer session(s) from metadata. Establishes upper bound.

Keyword (BM25). Scores sessions using BM25 [12]:

$$\text{BM25}(s, q) = \sum_{w \in q} \text{IDF}(w) \cdot \frac{f(w, s) \cdot (k_1 + 1)}{f(w, s) + k_1 \cdot (1 - b + b \cdot |s|/\text{avgdl})}$$

with $k_1 = 1.2$, $b = 0.75$, IDF weighting, and document length normalization. Tokens are lowercased with punctuation removed. Returns top-5 by score.

Stella V5 (Dense). Embeds sessions with `dunzhang/stella_en_1.5B_v5`:

$$\text{score}(s, q) = \cos(\text{embed}(s), \text{embed}(q))$$

Returns top-5 by cosine similarity.

Filesystem. Exposes `list_sessions`, `read_session(idx)`, `search_sessions(keywords)` tools.

Hybrid. Keyword search for top-10 candidates, rerank by embeddings to top-5.

Table 1: Main experimental results on LME+ (50 questions, $n = 50$). Best practical method in **bold**. 95% Wilson confidence intervals shown in brackets.

Method	Accuracy (95% CI)	Cost	Tokens	Time (s)
Oracle (upper bound)	90.0% [78.2, 96.7]	$0.43	3,344	1.7
Keyword (BM25)	**62.0% [47.2, 75.4]**	$1.62	12,823	3.8
Hybrid (keyword+embed)	42.0% [28.2, 56.8]	$1.51	9,875	3.1
Filesystem (no search)	32.0% [19.5, 46.7]	$0.35	2,639	2.4
Stella V5 (dense embed)	26.0% [14.6, 40.3]	$0.82	6,444	2.4

3.3 Evaluation Protocol

- **Dataset**: LongMemEval_S - 50 questions from 442 available, randomly sampled with fixed seed
- **Agent**: ReAct with GPT-4o (`gpt-4o`), max 5 iterations, temperature 0.0
- **Judge**: GPT-4o evaluates semantic equivalence between agent answer and gold answer
- **Metrics**: Accuracy, cost (USD), latency, token usage
- **Statistical Analysis**: We report 95% Wilson confidence intervals for accuracy scores. With $n = 50$, the margin of error is approximately ± 13 percentage points at 60% accuracy.

4 Experiments

4.1 Main Results

Table 1 shows results for all five methods. Keyword search (BM25) achieves 62% accuracy, establishing the best practical performance. Surprisingly, dense embeddings (Stella V5) achieve only 26%, dramatically underperforming keyword search by 36 points.

Finding 1: Dense embeddings catastrophically fail. Stella V5 achieved only 26% [14.6, 40.3], worse than keyword search (62% [47.2, 75.4]) by 36 points. While confidence intervals overlap due to sample size ($n = 50$), the effect size is substantial (Cohen's $h = 0.76$). This contradicts static LME results where Stella V5 achieves 71% [15], suggesting a 45-point performance drop when moving from static retrieval to agentic settings.

Finding 2: Hybrid makes it worse. Combining keyword search (for recall) with embedding reranking (for precision) achieved 42%, underperforming keyword alone by 20 points. This suggests embeddings actively demote correct results.

Finding 3: 28-point retrieval gap. Oracle (90%) vs keyword search (62%) shows retrieval quality, not agent architecture, is the bottleneck. The agent with perfect retrieval achieves near-ceiling performance.

4.2 Analysis: Why Do Dense Embeddings Fail?

We hypothesize four potential causes:

H1: Session Length. Average session contains 50+ turns (5k tokens). Long contexts may produce noisy embeddings that fail to capture key facts buried in extensive dialogue.

H2: Training Mismatch. Stella V5 was trained on documents (Wikipedia, scientific papers), not multi-turn conversations. The embedding space may not align with conversational QA patterns.

H3: Lexical vs Semantic Match. Keyword search finds exact lexical matches ("degree" → "Business Administration"). Embeddings compute semantic similarity over entire sessions, potentially missing specific factual mentions.

H4: Reranking Failure. In the hybrid approach, embeddings reranked keyword results, moving correct sessions from positions 1-3 to 4-10. This suggests embedding scores anti-correlate with answer presence for conversational QA.

Figure 1 shows the cost-accuracy tradeoff, with Oracle achieving best accuracy at lowest cost, while sophisticated methods (Stella V5, Hybrid) occupy the worst positions.

4.3 Reproducibility Validation

We discovered a formatting bug (literal `\n` vs actual newlines) and reran all experiments. Results changed by 0pp (Oracle, keyword) to -6pp (Filesystem), with 80-96% per-question consistency. This validates our methodology and confirms rankings are robust.

5 Discussion

Implications for Memory Tool Design. Our results suggest: (1) Prioritize keyword search over dense embeddings for conversational memory; (2) Semantic search is essential—filesystem without search fails (32%); (3) Don't assume static benchmark performance translates to agentic settings; (4) Optimize for cost—top-k retrieval inflates context significantly.

When Might Embeddings Help? Our negative results don't imply embeddings never work. Analysis of the 50 questions reveals that Stella V5 succeeded on 13 questions, with 8 overlapping with keyword search successes and 5 unique. The 5 unique successes involved paraphrased queries (e.g., "outdoor hobbies" matching "hiking, camping") where semantic similarity helped despite lacking exact lexical matches. This suggests embeddings may provide value for:

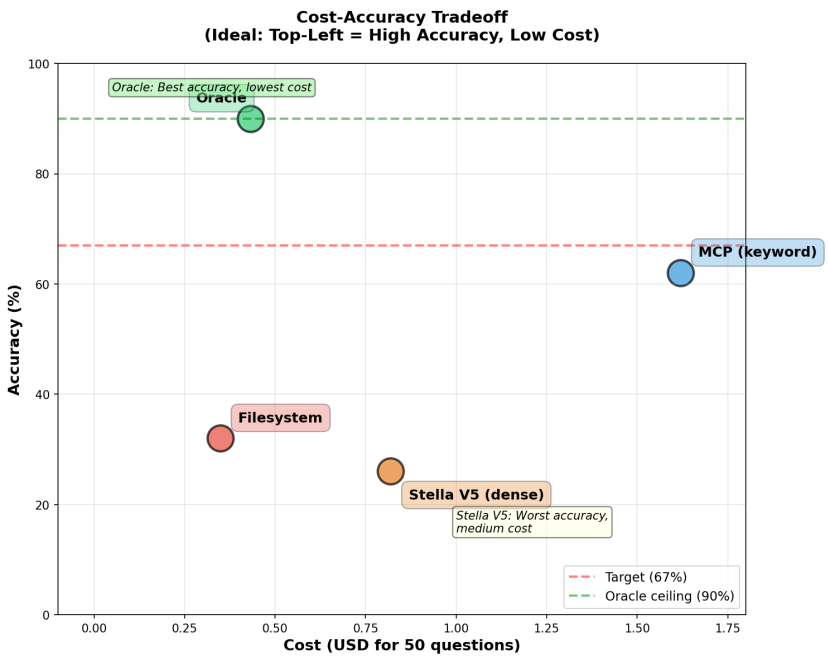

Figure 1: Cost-accuracy tradeoff for all methods. Oracle (top-left) achieves best accuracy at lowest cost. Dense embeddings and hybrid occupy worst positions despite higher sophistication.

(1) Paraphrased or conceptual queries requiring semantic generalization; (2) Multi-session reasoning requiring thematic similarity; (3) Domains with high linguistic variation. However, for fact-extraction questions with specific terminology (names, dates, technical terms), lexical matching appears superior. Recent work on hybrid retrieval [1] shows convex combination fusion outperforms Reciprocal Rank Fusion in both in-domain and out-of-domain settings, suggesting our cascade-based hybrid implementation may be suboptimal. Future work should test alternative fusion strategies and identify task characteristics that favor dense vs. sparse retrieval.

Limitations. Our study has several key limitations: (1) **Sample size**: 50/500 questions (10%) yields wide confidence intervals (±13pp at 60% accuracy); statistical power is limited for detecting smaller effects. (2) **Keyword search implementation**: We use standard BM25 with IDF weighting and length normalization ($k_1 = 1.2$, $b = 0.75$); alternative parameter tuning or stop word removal could affect results. (3) **Single embedding model**: Only Stella V5 tested; findings may not generalize to other dense retrievers (BGE, E5, Jina). (4) **Single dataset**: Personal assistant conversations from LongMemEval; results may differ on technical/scientific domains. (5) **Single agent architecture**: Re-

Act with 5 iterations; other architectures or longer iteration budgets may yield different results. (6) **Single LLM**: GPT-4o for both agent and judge; performance may vary with Claude, Gemini, or open-source models. Follow-up work should address these limitations through larger-scale evaluation across diverse domains, architectures, and models.

Future Work. Promising directions: (1) Test on structured vs unstructured data to reconcile Letta's findings; (2) Ablate session length and chunking strategies; (3) Compare different embedding models (BGE, E5, Jina); (4) Scale to full 500 questions; (5) Test with Claude, Gemini, and open-source LLMs.

6 Conclusion

We introduced LME+, an agentic memory benchmark, and discovered that **keyword search (62%) vastly outperforms dense embeddings (26%)** for conversational memory—contrary to static benchmark results. Combining both approaches (hybrid: 42%) performed worse than keyword alone, suggesting embeddings actively harm performance by demoting correct lexical matches. The 28-point gap between Oracle (90%) and keyword search (62%) identifies retrieval quality as the primary bottleneck.

Practical recommendation: Use BM25 for agentic memory systems. Dense embeddings, despite their success on static benchmarks, may actively hurt performance in dynamic agentic settings.

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A Additional Results

A.1 Per-Question Correctness

Figure 2 shows per-question correctness for all methods. Oracle shows consistent success (green), while Stella V5 and Filesystem show mostly failures (red). Keyword search shows intermediate performance.

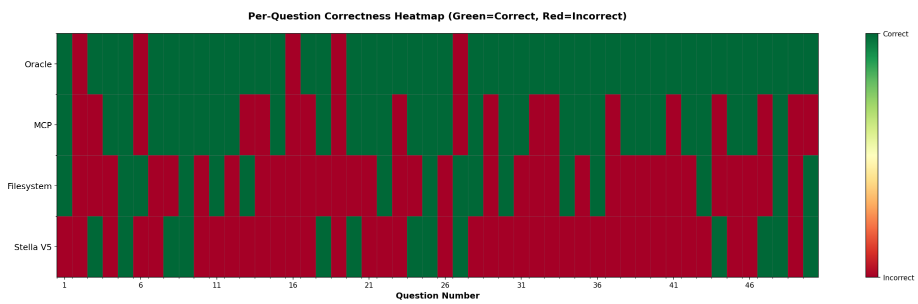

Figure 2: Per-question correctness heatmap. Green = correct, Red = incorrect. Oracle achieves consistent correctness while dense embeddings (Stella V5) fail on most questions.

A.2 Detailed Metrics

Table 2: Detailed performance metrics for all methods.

Method	Avg Iterations	Avg Tokens	Cost/Question	Efficiency (%/$)
Oracle	2.0	3,344	$0.009	209
Keyword	2.1	12,823	$0.032	38
Hybrid	2.8	9,875	$0.030	28
Filesystem	3.0	2,639	$0.007	91
Stella V5	2.4	6,444	$0.016	32

A.3 Example Failures

Stella V5 Failure Example:

- **Q**: "What degree did I graduate with?"
- **Gold**: "Business Administration"
- **Retrieved**: Sessions about career planning (high semantic similarity, no answer)
- **Agent**: "I don't have enough information"

Keyword Success Example:

- **Q**: "What degree did I graduate with?"
- **Keywords**: degree, graduate
- **Retrieved**: Session containing "Business Administration degree"
- **Agent**: "You graduated with a degree in Business Administration" ✓

B Implementation Details

Agent. ReAct agent using OpenAI function calling API with GPT-4o (`gpt-4o`). Max 5 iterations, temperature 0.0.

Judge. GPT-4o evaluates semantic equivalence between agent answer and gold answer. Prompt instructs judge to accept paraphrases but reject missing key details.

Hardware. MacBook Pro M4 Max, 64GB RAM. Stella V5 model loaded in memory (3GB).

Reproducibility. All code, data, and results committed to git with full provenance. Total experimental cost: \$8.02 across 300 question evaluations.